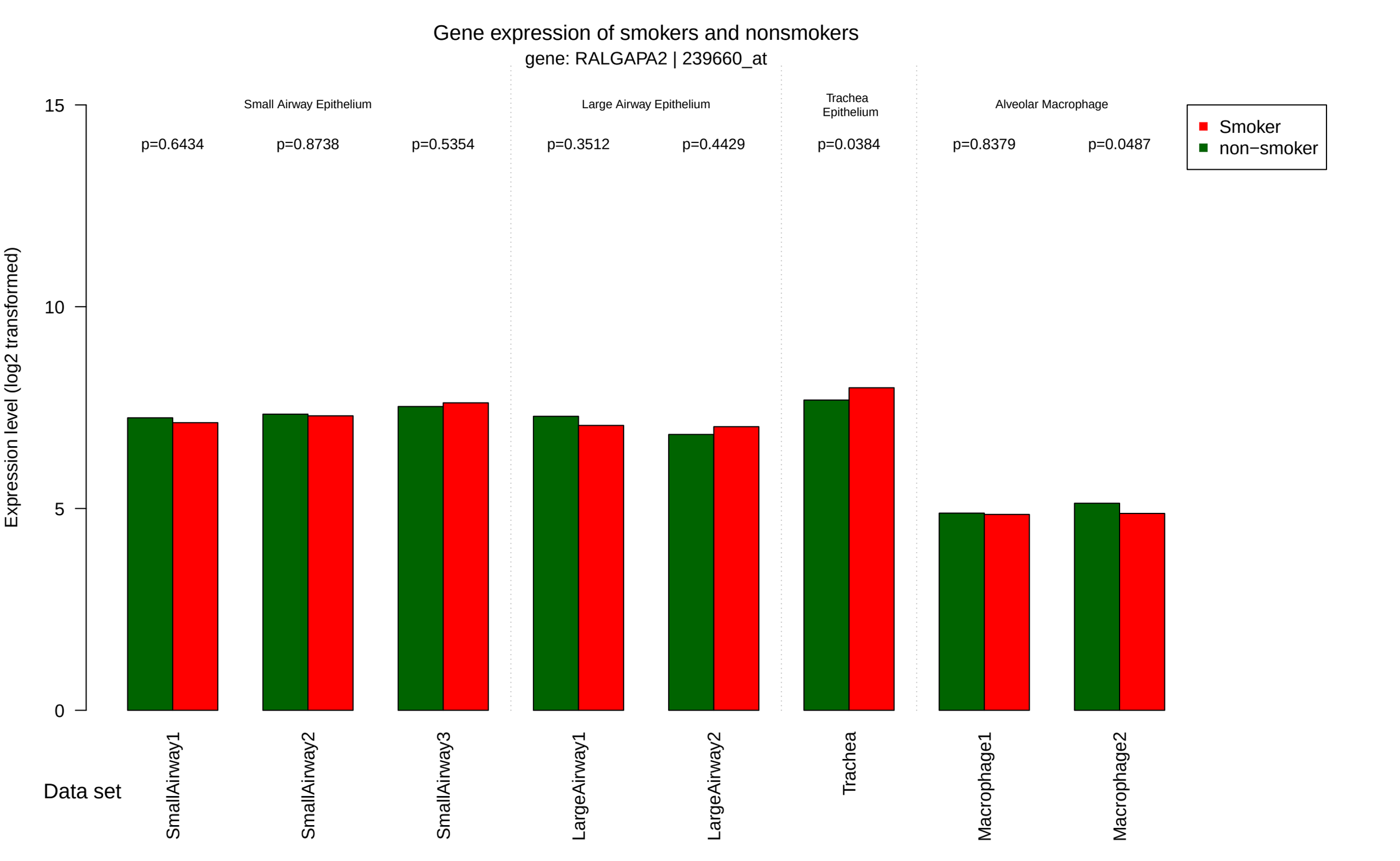
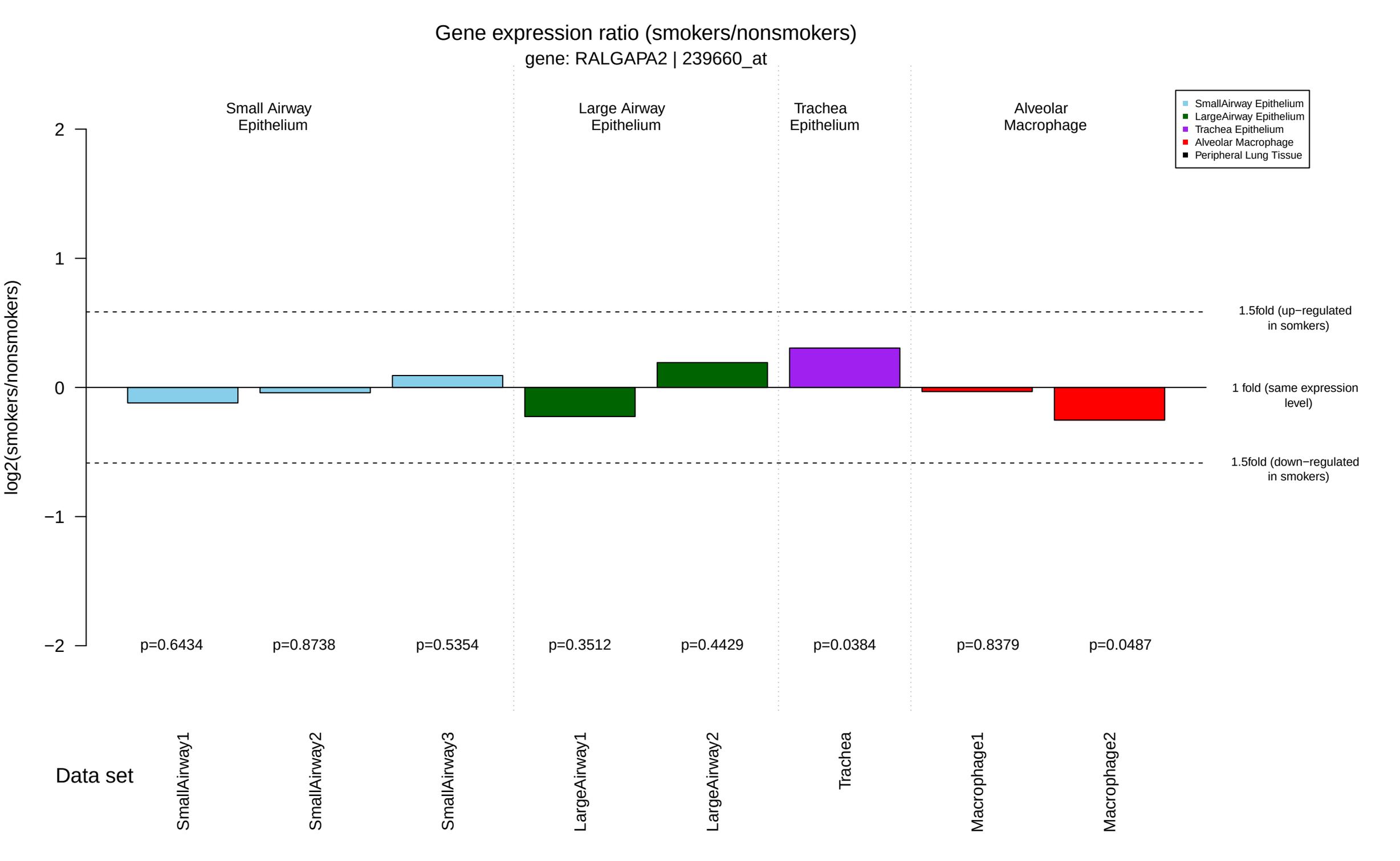
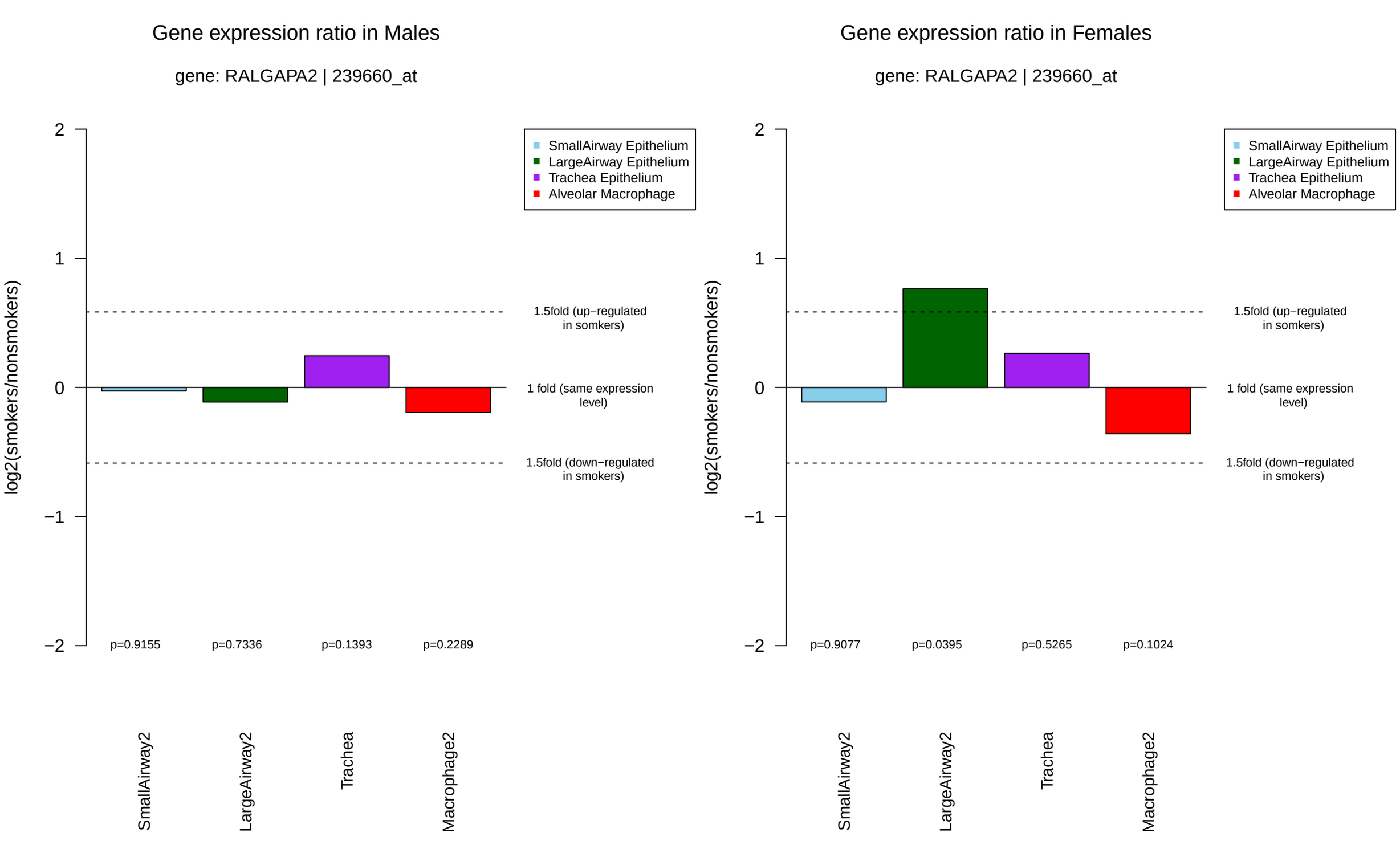
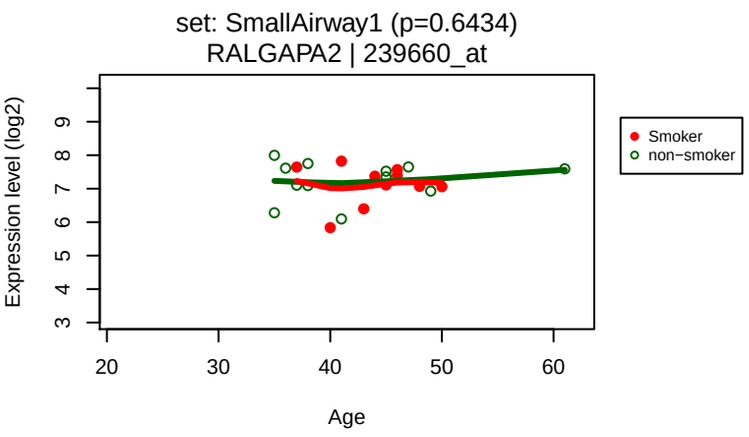
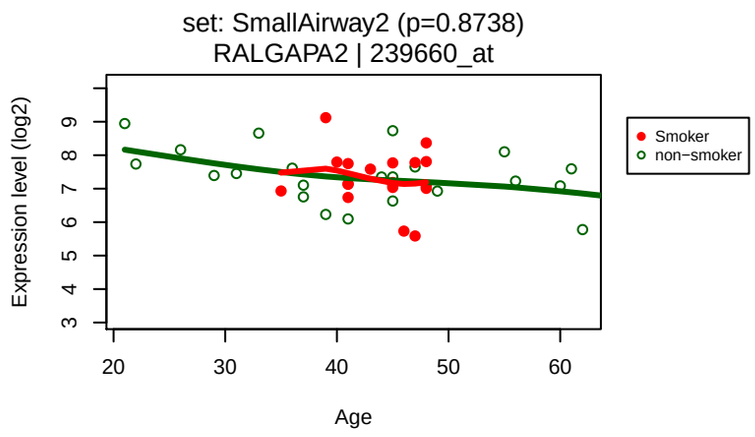
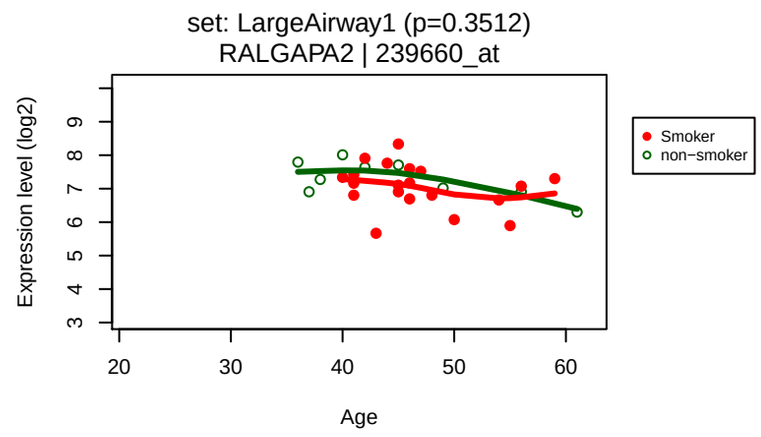

Gene expression of smokers and nonsmokers
gene: RALGAPA2 | 239660_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6434
p=0.8738
p=0.5354
p=0.3512
p=0.4429
p=0.0384
p=0.8379
p=0.0487
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

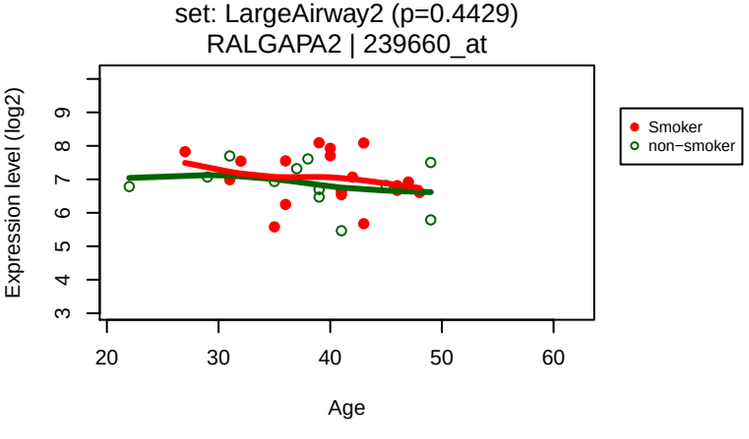

Gene expression ratio (smokers/nonsmokers)
gene: RALGAPA2 | 239660_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6434
p=0.8738
p=0.5354
p=0.3512
p=0.4429
p=0.0384
p=0.8379
p=0.0487
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

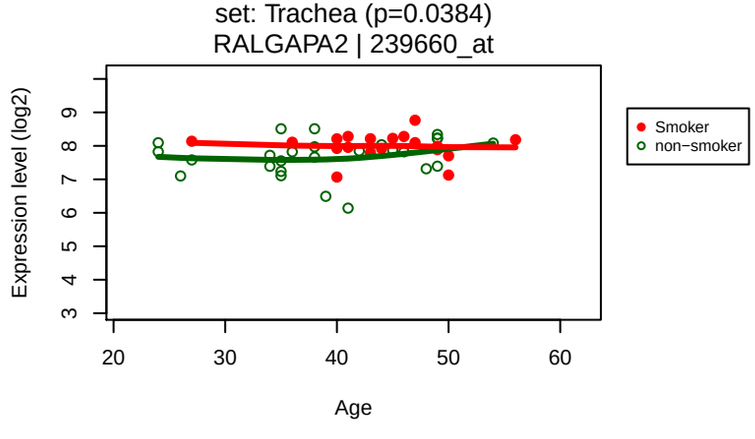

Gene expression ratio in Males
gene: RALGAPA2 | 239660_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9155
p=0.7336
p=0.1393
p=0.2289
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RALGAPA2 | 239660_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9077
p=0.0395
p=0.5265
p=0.1024
SmallAirway2
LargeAirway2
Trachea
Macrophage2

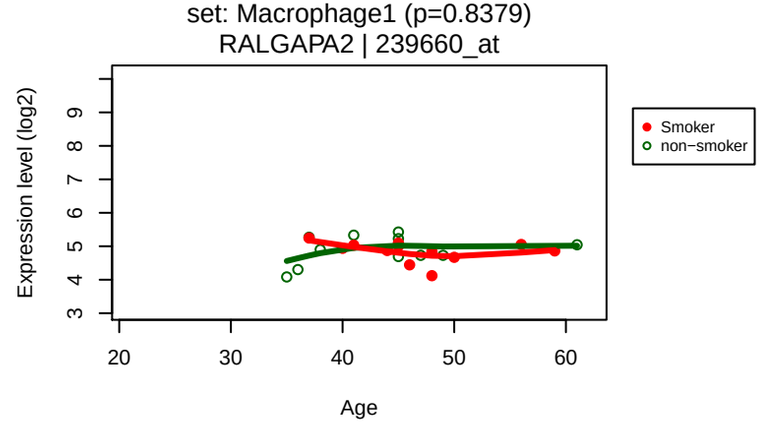

set: SmallAirway1 (p=0.6434)
RALGAPA2 | 239660_at
Expression level (log2)
Age
Smoker
non−smoker

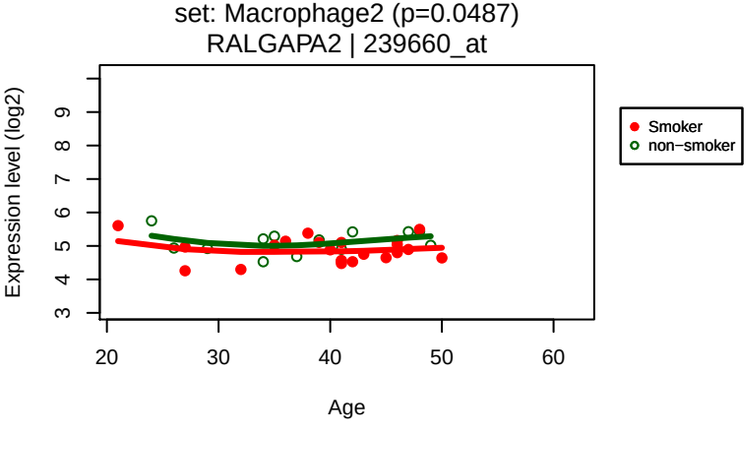

set: SmallAirway2 (p=0.8738)
RALGAPA2 | 239660_at
Expression level (log2)
Age
Smoker
non−smoker

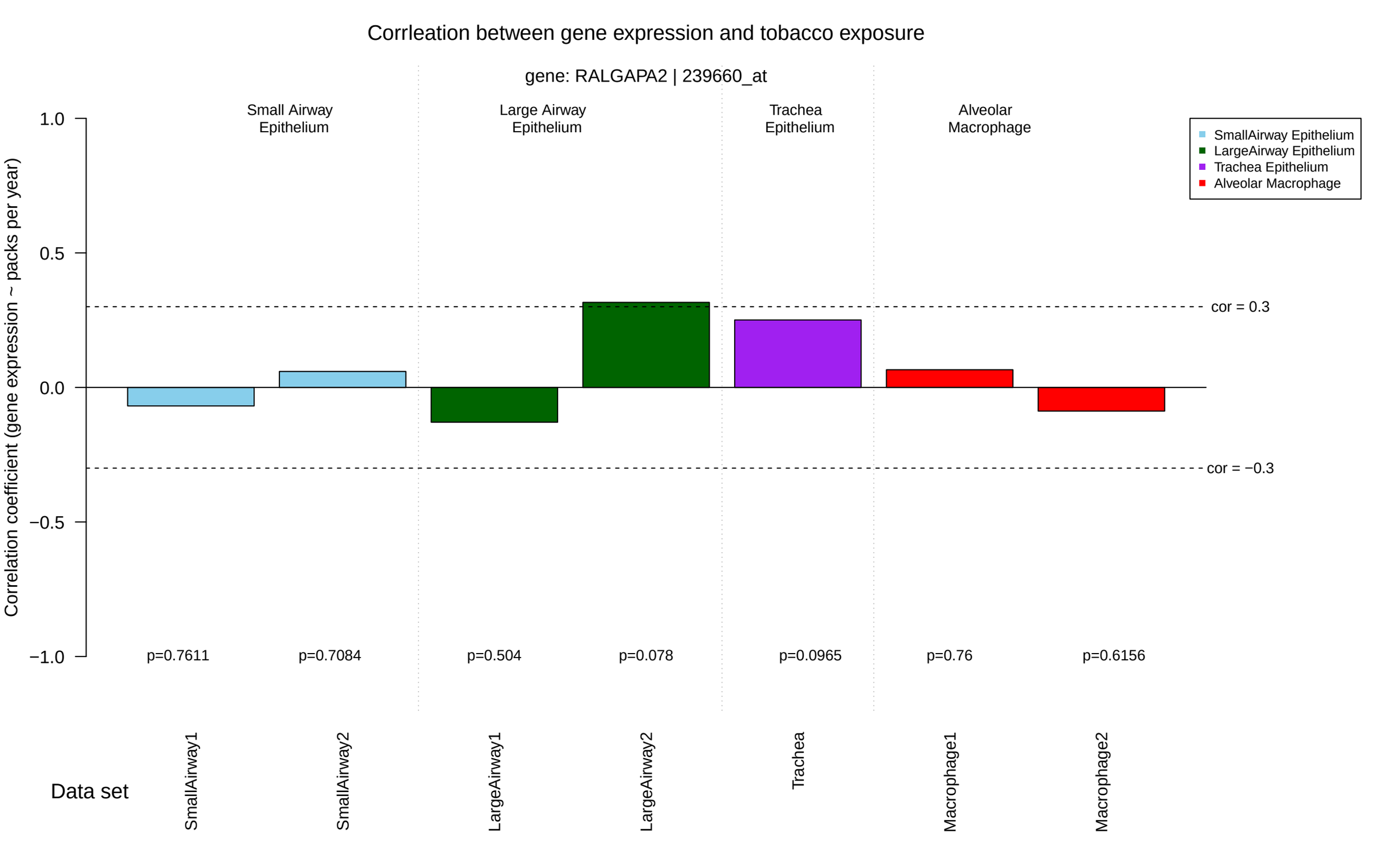

set: LargeAirway1 (p=0.3512)
RALGAPA2 | 239660_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4429)
RALGAPA2 | 239660_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0384)
RALGAPA2 | 239660_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8379)
RALGAPA2 | 239660_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0487)
RALGAPA2 | 239660_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RALGAPA2 | 239660_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7611
p=0.7084
p=0.504
p=0.078
p=0.0965
p=0.76
p=0.6156
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2